Class 9 Science: Chemistry

NCERT syllabus 2026-27 par based

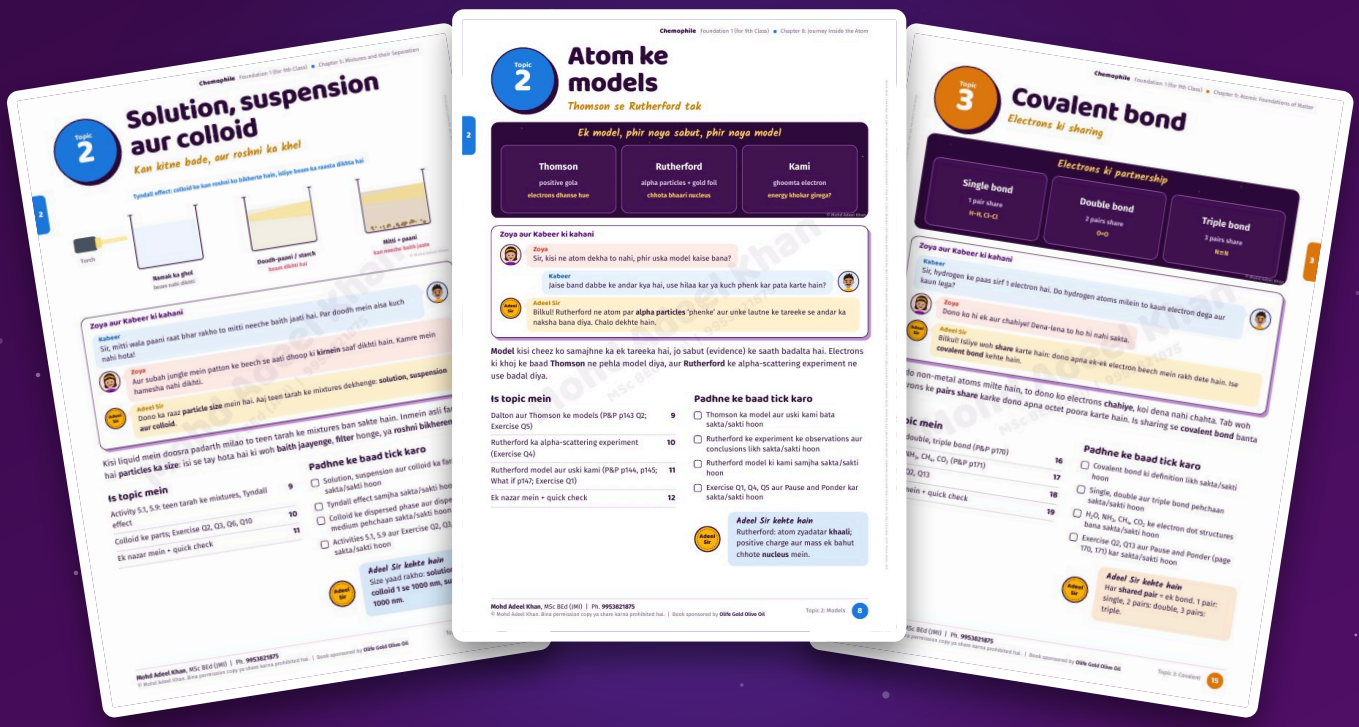
Chemophile

Foundation 1 (for 9th Class)

FREE SAMPLE
har chapter se ek topic

Har chapter ki ek jhalak

NCERT Class 9 Science ke saare 3 chemistry chapters, easy Hinglish mein, Zoya aur Kabeer ke saath



5 Mixtures and their Separation
8 Journey Inside the Atom
9 Atomic Foundations of Matter

121 colourful pages | **57** figures aur diagrams | **28** solved examples | **310+** practice questions | **3** chapter tests (40 marks)

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Chemophile kya hai?

Chemophile Foundation 1 (for 9th Class) NCERT Class 9 Chemistry ke saare **3 chapters** ki colourful booklets hain. Explanations **modern Hinglish** mein, technical terms aur exam answers **English** mein. Har topic ek story ki tarah chalta hai: Adeel Sir ka example, visual, concept, exam wala answer, aur end mein practice.

3 chapters	**121** colourful pages	**57** figures aur diagrams	**28** solved examples	**310+** practice questions

Har page par kya milega

Zoya aur Kabeer ki kahani	Har topic ek chhoti si story se shuru hota hai.
NCERT Activity	Activity kaise karein, kya dikhta hai, aur kyun.
Galti mat karna	Common mistakes jo marks cut karwati hain.
Mystery clue	Har chapter ek mystery; clues collect karke solve karo.
Adeel Sir ka example	Har concept ghar ki kisi cheez se samjhaya gaya hai.
Exam mein aise likho	Board exam ke liye English answer, ruled lines par.
Aage kaam aayega	Thoda extra jo Class 11 aur 12 mein kaam aayega.
Ghar par karo	Ghar ki cheezon se chhoti, safe activities.

Har chapter ke andar

- ✓ Har topic: Zoya-Kabeer ki story, example, visual, phir concept
- ✓ NCERT ke saare exercise questions solved
- ✓ Har topic ke end mein recap, quick check aur answers
- ✓ Har chapter ke end mein revision aur 40 marks ka test
- ✓ NCERT ki saari activities step by step
- ✓ Exam mein aise likho: board ke liye English answers
- ✓ Har chapter ka mystery, crossword aur Galti dhundo
- ✓ Bonus booklet, revision sheets aur mock tests alag se

Is sample mein kya hai?

Har chapter se **ek topic** ke do asli pages: topic ka pehla page aur uska pehla lesson page. Phir Chapter 9 ke revision ke do pages (fact sheet aur chapter test ka pehla page). Yeh pages bilkul waise hi hain jaise poori book mein: **121 pages** isi style mein.

Adeel Sir kehte hain

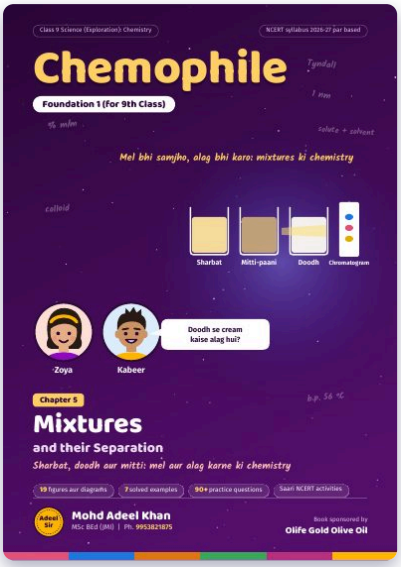
Chemistry ratne ka subject nahi, samajhne ka subject hai. Jab concept dikh jaata hai, to formula khud yaad ho jaata hai.

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Saare 3 chapters ek nazar mein

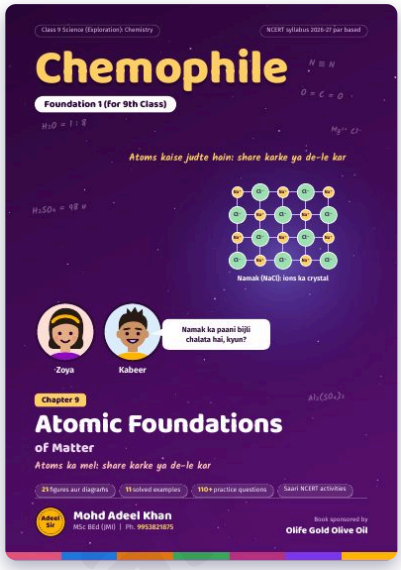
Har chapter ki apni booklet hai. Neeche har booklet ka cover, uske pages aur is sample mein shaamil topic.



Chapter 5 Mixtures and their Separation
36 pages · Sample: Solution, suspension aur colloid



Chapter 8 Journey Inside the Atom
40 pages · Sample: Atom ke models: Thomson se Rutherford



Chapter 9 Atomic Foundations of Matter
45 pages · Sample: Covalent bond: electrons ki sharing

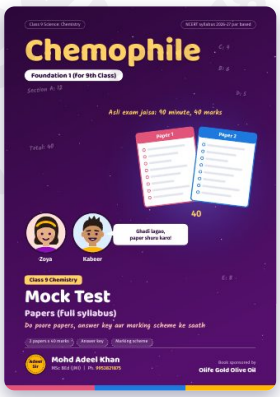
Saath mein extra booklets



Bonus: Mole Concept
28 pages · Class 11 ki taiyaari



Quick Revision Sheets
Har chapter ek page



Mock Test Papers
2 papers · answer key



Complete Volume
161 pages · sab ek saath

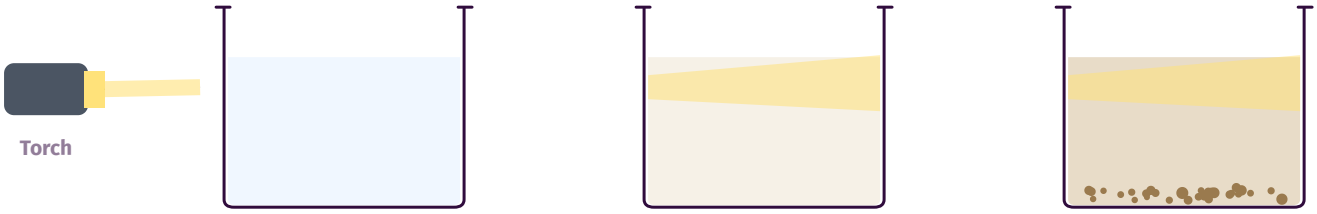
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Topic 2

Solution, suspension aur colloid

Kan kitne bade, aur roshni ka khel

Tyndall effect: colloid ke kan roshni ko bikherte hain, isliye beam ka raasta dikhta hai



Namak ka ghol
beam nahi dikhti

Doodh-paani / starch
beam dikhti hai

Mitti + paani
kan neeche baith jaate

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Zoya aur Kabeer ki kahani

Kabeer
Sir, mitti wala paani raat bhar rakho to mitti neeche baith jaati hai. Par doodh mein aisa kuch nahi hota!

Zoya
Aur subah jungle mein patton ke beech se aati dhoop ki **kirnein** saaf dikhti hain. Kamre mein hamesha nahi dikhti.

Adeel Sir
Dono ka raaz **particle size** mein hai. Aaj teen tarah ke mixtures dekhenge: **solution, suspension aur colloid**.

Kisi liquid mein doosra padarth milao to teen tarah ke mixtures ban sakte hain. Inmein asli farak hai **particles ka size**: isi se tay hota hai ki woh **baith jaayenge**, **filter** honge, ya **roshni bikherenge**.

Is topic mein

Activity 5.1, 5.9: teen tarah ke mixtures, Tyndall effect	**9**
Colloid ke parts; Exercise Q2, Q3, Q6, Q10	**10**
Ek nazar mein + quick check	**11**

Padhne ke baad tick karo

- [ ] Solution, suspension aur colloid ka farak bata sakta/sakti hoon
- [ ] Tyndall effect samjha sakta/sakti hoon
- [ ] Colloid ke dispersed phase aur dispersion medium pehchaan sakta/sakti hoon
- [ ] Activities 5.1, 5.9 aur Exercise Q2, Q3, Q6, Q10 kar sakta/sakti hoon

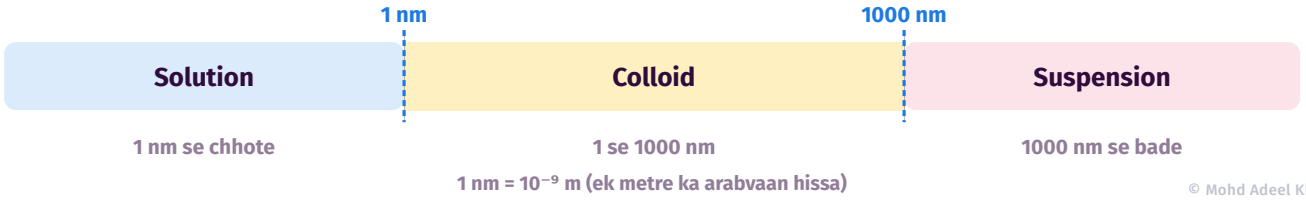
Adeel Sir kehte hain
Size yaad rakho: **solution < 1 nm, colloid 1 se 1000 nm, suspension > 1000 nm.**

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Activity 5.1

Samajh aaya? Rate karo ☆ ☆ ☆

Solution, suspension aur colloid



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NCERT Activity 5.1 · kaun solution, kaun suspension, kaun colloid?

Kya chahiye

Paani, namak, cheeni, chalk powder, mitti, doodh ya starch, filter paper, torch

Kya karna hai

1. Alag glasses mein paani lekar har cheez thodi milao, hilao.
2. Dekho: ghol saaf (transparent) hai ya dhundhla? Kuch der rakhne par kuch neeche baitha?
3. Har ek ko filter karo, aur andhere mein torch ki roshni guzaaro.

Kya dikhta hai

Namak, cheeni: **saaf**, kuch nahi baitha, filter se kuch nahi ruka, beam nahi dikhi (**solution**). Chalk, mitti: **dhundhla**, kan **baith gaye**, filter par ruke (**suspension**). Doodh, starch: dhundhla, baithe nahi, filter se guzar gaye, par **beam dikhi** (**colloid**).

Property (Activity 5.9)	Solution	Colloid	Suspension
Particle size	< 1 nm	1 se 1000 nm	> 1000 nm
Homogeneous?	Haan	Dikhne mein haan, asal mein nahi	Nahi
Tyndall effect	Nahi	Haan	Haan (kan bade hain)
Rakhne par baithte hain?	Nahi	Nahi (stable)	Haan
Filter paper se alag?	Nahi	Nahi	Haan

Tyndall effect: colloid ke kan roshni ko **bikher (scatter)** dete hain, isliye uske andar se guzarti roshni ka **raasta dikhta** hai. Solution ke kan itne chhote hain ki roshni nahi bikherte. **Solution** mein ghulne wala padarth **solute** aur ghulaane wala **solvent** kehlaata hai (namak ke ghol mein namak solute, paani solvent).

Exam mein aise likho

Think it Over (page 72). Why does muddy water settle on standing but milk does not?

Muddy water is a suspension: its particles are large (more than 1000 nm) and settle down under gravity.

Milk is a colloid: its particles are very small (1 to 1000 nm) and remain spread throughout the water, so they do not settle on standing.

Adeel Sir ka solved example 2.1

Exercise Q2 [paraphrased]. Which of these show the Tyndall effect: air with dust, copper sulphate in water, starch in water, acetone in water?

Colloid / bade kan Hawa + dhool aur starch + paani: kan roshni bikherte hain

Solution $CuSO_4$ + paani aur acetone + paani: true solutions

Air with dust and starch in water show the Tyndall effect

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Topic 2

Atom ke models

Thomson se Rutherford tak

Ek model, phir naya sabut, phir naya model

Thomson	Rutherford	Kami
positive gola	alpha particles + gold foil	ghoomta electron
electrons dhanse hue	chhota bhaari nucleus	energy khokar girega?

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Zoya aur Kabeer ki kahani



Zoya
Sir, kisi ne atom dekha to nahi, phir uska model kaise bana?

Kabeer
Jaise band dabbe ke andar kya hai, use hilaa kar ya kuch phenk kar pata karte hain?



Adeel Sir

Adeel Sir
Bilkul! Rutherford ne atom par **alpha particles** 'phenke' aur unke lautne ke tareeke se andar ka naksha bana diya. Chalo dekhte hain.

Model kisi cheez ko samajhne ka ek tareeka hai, jo sabut (evidence) ke saath badalta hai. Electrons ki khoj ke baad **Thomson** ne pehla model diya, aur **Rutherford** ke alpha-scattering experiment ne use badal diya.

Is topic mein

Dalton aur Thomson ke models (P&P p143 Q2; Exercise Q5)	9
Rutherford ka alpha-scattering experiment (Exercise Q4)	10
Rutherford model aur uski kami (P&P p144, p145; What if p147; Exercise Q1)	11
Ek nazar mein + quick check	12

Padhne ke baad tick karo

- [ ] Thomson ka model aur uski kami bata sakta/sakti hoon
- [ ] Rutherford ke experiment ke observations aur conclusions likh sakta/sakti hoon
- [ ] Rutherford model ki kami samjha sakta/sakti hoon
- [ ] Exercise Q1, Q4, Q5 aur Pause and Ponder kar sakta/sakti hoon

Adeel Sir

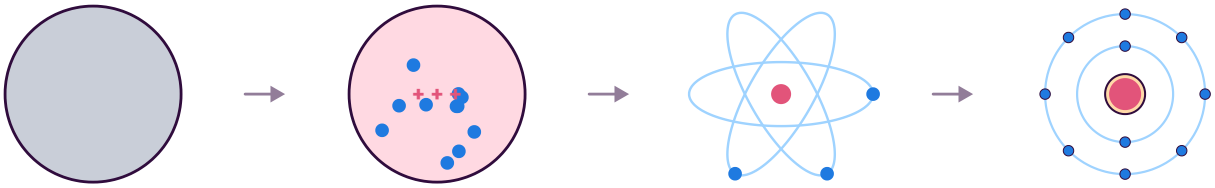
Adeel Sir kehte hain
Rutherford: atom zyadatar **khaali**; positive charge aur mass ek bahut chhote **nucleus** mein.

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Atomic models

Samajh aaya? Rate karo ☆ ☆ ☆

Dalton se Thomson tak



Dalton (1808)
thos, na tootne wala gola

Thomson (1904)
positive gola, electrons dhanse

Rutherford (1911)
chhota bhaari nucleus

Bohr (1913)
fixed orbits / shells

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Dalton (1808): atom ek thos, **na tootne wala** gola. Electron mile to yeh idea adhoora ho gaya.

Thomson ka model (plum pudding): atom ek **positive charge wala gola** hai jismein **electrons** jagah-jagah dhanse hue hain, jaise **tarbooz mein beej** ya pudding mein kishmish. Kul positive charge = kul negative charge, isliye atom neutral.

Thomson model ne kya samjhaya	Kya nahi samjha paaya
Atom neutral kyun hai; electrons atom ka hissa hain	Alpha particles ka **laut aana** (Rutherford ka experiment); atom ke andar charge kaise bikhra hai

Exam mein aise likho

Pause and Ponder (page 143) [based on]. An orange or a lemon is sometimes used to explain Thomson's model. How far does this analogy work?

It works partly: the pulp is like the uniform positive sphere and the seeds are like electrons inside it. But it fails because the seeds are not spread evenly, the fruit carries no charge, and the seeds are not tiny compared with the fruit as electrons are compared with the atom.

Adeel Sir ka solved example 2.1

Exercise Q5 [paraphrased]. Arrange the models of the atom in the order in which they were proposed: Bohr, Thomson, Dalton, Rutherford.

Yaad karo Dalton 1808, Thomson 1904, Rutherford 1911, Bohr 1913

Kyun yahi order Har naya model pichhle ki kami door karne aaya

Dalton → Thomson → Rutherford → Bohr

⚠ **Galti mat karna.** Thomson model mein **nucleus nahi** hai. Positive charge poore gole mein **bikhra** hai. Nucleus Rutherford ne diya.

✓ **Ghar par karo**

Model banao: gol aata ki loi (positive gola) mein 8 rajma ke daane (electrons) dabao. Yeh Thomson ka model hai. Ab socho: Rutherford ka model banane ke liye loi ko kya karna padega?

Ab tumhari baari *answers topic ke end mein*

1. What carries the positive charge in Thomson's model?

Answer

2. Why is Thomson's atom neutral?

Answer

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Topic 3

Covalent bond

Electrons ki sharing

Electrons ki partnership

Single bond	Double bond	Triple bond
1 pair share	2 pairs share	3 pairs share
H–H, Cl–Cl	O=O	N≡N

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Zoya aur Kabeer ki kahani



Kabeer
Sir, hydrogen ke paas sirf 1 electron hai. Do hydrogen atoms milein to kaun electron dega aur kaun lega?



Zoya
Dono ko hi ek aur chahiye! Dena-lena to ho hi nahi sakta.



Adeel Sir
Bilkul! Isliye woh **share** karte hain: dono apna ek-ek electron beech mein rakh dete hain. Ise **covalent bond** kehte hain.

Jab do non-metal atoms milte hain, to dono ko electrons **chahiye**, koi dena nahi chahta. Tab woh electrons ke **pairs share** karke dono apna octet poora karte hain. Is sharing se **covalent bond** banta hai.

Is topic mein

Single, double, triple bond (P&P p170)	16
F_2, H_2O, NH_3, CH_4, CO_2 (P&P p171)	17
Exercise Q2, Q13	18
Ek nazar mein + quick check	19

Padhne ke baad tick karo

- ☐ Covalent bond ki definition likh sakta/sakti hoon
- ☐ Single, double aur triple bond pehchaan sakta/sakti hoon
- ☐ H_2O, NH_3, CH_4, CO_2 ke electron dot structures bana sakta/sakti hoon
- ☐ Exercise Q2, Q13 aur Pause and Ponder (page 170, 171) kar sakta/sakti hoon



Adeel Sir kehte hain
Har **shared pair** = ek bond. 1 pair: single, 2 pairs: double, 3 pairs: triple.

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Covalent

Samajh aaya? Rate karo ☆ ☆ ☆

Covalent bond: electrons ki sharing

Adeel Sir ka example

Do dost ek hi cycle share karte hain, aur dono kehte hain 'meri cycle'. Covalent bond mein bhi do atoms electrons ka ek **pair** share karte hain, aur **dono** us pair ko apne octet mein ginte hain.

Covalent bond: do atoms ke beech **electrons ki sharing** se bana bond. Zyadatar yeh **non-metals** ke beech banta hai, aur isse **molecule** banta hai. **Electron dot structure** mein sirf bahar wale shell ke electrons dikhaate hain; alag atoms ke electrons alag nishaan (•, ×) se.

	H_2: single bond	O_2: double bond	N_2: triple bond
Dot:	H •× H	O ×× O	:N ××× N:
Line:	H—H	O=O	N≡N
	1 shared pair	2 shared pairs	3 shared pairs

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Molecule	Atom ki configuration	Kya hota hai	Bond
H_2	H: 1	Dono ek-ek electron share karte hain: dono ke paas **duplet**	Single (H–H)
O_2	O: 2, 6 (2 chahiye)	Dono 2-2 electrons share karte hain: **2 shared pairs**, dono ka octet	Double (O=O)
N_2	N: 2, 5 (3 chahiye)	Dono 3-3 electrons share karte hain: **3 shared pairs**	Triple (N≡N)

Exam mein aise likho

Pause and Ponder (page 170) [based on]. Explain the formation of a nitrogen molecule. Why does it have a triple bond?

A nitrogen atom (2, 5) has five outer electrons and needs three more to complete its octet. Two nitrogen atoms each share three electrons, forming three shared pairs between them. So the N_2 molecule has a triple covalent bond (N≡N), and each atom also keeps one lone pair.

⚠ **Galti mat karna.** Bond ki ek line (–) ek electron nahi, ek **shared pair** yaani **2 electrons** dikhati hai. Double bond = 4 electrons.

Ab tumhari baari *answers topic ke end mein*

1. How many electrons are shared in an O_2 molecule?

   Answer

2. Which type of bond is present in Cl_2?

   Answer

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Revision

Samajh aaya? Rate karo ☆ ☆ ☆

Poora chapter ek page par

Law	Kya kehta hai	Example
Conservation of mass (Lavoisier)	Chemical reaction mein mass na banta hai, na khatam hota hai	$BaCl_2 + Na_2SO_4$: mass pehle = baad mein
Constant proportions (Proust)	Compound mein elements hamesha fixed mass ratio mein	Paani: H : O = 1 : 8

Term	Matlab	Example
Atomicity	Ek molecule mein atoms ki sankhya	He 1, O_2 2, O_3 3, P_4 4, S_8 8
Cation / anion	Electron dekar bana + ion / lekar bana − ion	Na^+, Mg^{2+} / Cl^-, O^{2-}
Covalent bond	Electron pairs ki **sharing** (non-metals)	H–H, O=O, N≡N, H_2O
Ionic bond	Electrons ka **transfer**, ions ka kheenchav (metal + non-metal)	NaCl, $MgCl_2$, CaO
Polyatomic ion	Atoms ka group jis par charge ho	NH_4^+, OH^-, SO_4^{2-}, PO_4^{3-}
Formula unit	Ionic compound mein ions ka simplest ratio	NaCl (1 : 1), $CaCl_2$ (1 : 2)
Molecular mass	Molecule ke atomic masses ka jod (u mein)	H_2O 18 u, CO_2 44 u

Yaad rakhne ke numbers	Value
1 u	C-12 atom ka 1/12
NaCl ka melting point	801 °C
H_2SO_4	98 u
NaCl	58.5 u

⚠ **Galti mat karna.** Formula mein subscript sirf **apne se pehle** wale symbol par lagta hai; bracket ke bahar wala **poore group** par: $Al_2(SO_4)_3$ mein 3 S aur 12 O.

Valency	Cations	Anions
1	Na^+, K^+, Ag^+, H^+, NH_4^+	Cl^-, Br^-, OH^-, NO_3^-, HCO_3^-
2	Mg^{2+}, Ca^{2+}, Zn^{2+}, Fe^{2+}, Cu^{2+}	O^{2-}, S^{2-}, CO_3^{2-}, SO_4^{2-}
3	Al^{3+}, Fe^{3+}	N^{3-}, PO_4^{3-}

Exam mein aise likho

Define: (a) law of conservation of mass (b) law of constant proportions (c) covalent bond (d) ionic bond (e) polyatomic ion (f) molecular mass.

(a) Mass can neither be created nor destroyed in a chemical reaction. (b) In a chemical compound, the elements are always present in a fixed ratio by mass. (c) A bond formed by the sharing of electron pairs between two atoms. (d) The electrostatic attraction between oppositely charged ions formed by the transfer of electrons. (e) A group of atoms that carries a charge and behaves as one unit, e.g., SO_4^{2-}. (f) The sum of the atomic masses of all the atoms in a molecule.

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Revision

Samajh aaya? Rate karo ☆ ☆ ☆

Test: Section A

Adeel Sir ke test rules

Book band karo. Poore test ke liye **90 minutes**, total **40 marks**. Section A: har question **1 mark**, sirf ek option.

Section A: ____ / 12

Q1 [Exam pattern] 10 g of $CaCO_3$ on heating gives 5.6 g of CaO and 4.4 g of CO_2. This shows the law of:

- ◯ constant proportions
- ◯ conservation of mass
- ◯ multiple proportions
- ◯ octaves

Q2 The mass ratio of hydrogen to oxygen in water is:

- ◯ 2 : 1
- ◯ 1 : 8
- ◯ 1 : 16
- ◯ 8 : 1

Q3 The atomicity of ozone is:

- ◯ 1
- ◯ 2
- ◯ 3
- ◯ 4

Q4 Which is a polyatomic ion?

- ◯ Cl^-
- ◯ NH_4^+
- ◯ Na^+
- ◯ O^{2-}

Q5 The number of shared electrons in N_2 is:

- ◯ 2
- ◯ 4
- ◯ 6
- ◯ 8

Q6 Which compound is covalent?

- ◯ NaCl
- ◯ MgO
- ◯ CH_4
- ◯ $CaCl_2$

Q7 The formula of aluminium sulphate is:

- ◯ $AlSO_4$
- ◯ Al_2SO_4
- ◯ $Al_2(SO_4)_3$
- ◯ $Al_3(SO_4)_2$

Q8 The number of electrons in a Cl^- ion is:

- ◯ 16
- ◯ 17
- ◯ 18
- ◯ 35

Q9 The formula unit mass of NaCl is:

- ◯ 23 u
- ◯ 35.5 u
- ◯ 58 u
- ◯ 58.5 u

Q10 Which of these conducts electricity?

- ◯ solid NaCl
- ◯ sugar solution
- ◯ NaCl solution
- ◯ kerosene

Assertion-Reason *(Q11 and Q12, 1 mark each)*

(a) Both A and R are true and R is the correct explanation of A. **(b)** Both A and R are true but R is not the correct explanation of A. **(c)** A is true but R is false. **(d)** A is false but R is true.

Q11 **Assertion (A):** The total mass in a closed flask stays the same when vinegar reacts with baking soda.
Reason (R): Atoms are neither created nor destroyed in a chemical reaction.

◯ (a) ◯ (b) ◯ (c) ◯ (d)

Q12 **Assertion (A):** Ionic compounds conduct electricity in the solid state.
Reason (R): Ionic compounds are made of ions.

◯ (a) ◯ (b) ◯ (c) ◯ (d)

Rough work

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Adeel Sir

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Is sample mein kaun se topics the

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8. Journey Inside the Atom	Topic 2: Atom ke models: Thomson se Rutherford	40 pages
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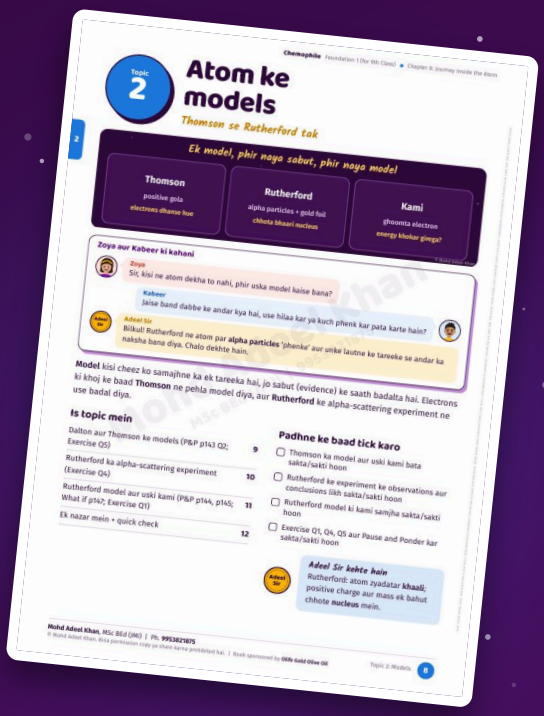
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